VR, AR and mixed reality (MR) can replicate the real environment using advanced processing, computing, sensing and display technology. One of the biggest benefits in a VR- and AR-enabled environment is the ability to innovate products and services in a completely new way.

Role of AI and ML in AR-VR integration

Artificial intelligence (AI) and machine learning (ML) technologies help reduce human error rates through AR and VR simulations of dangerous tasks and environments. India's skill development, training and learning space will create thousands of jobs opportunities in gaming, immersive advertising, creative media, movies, and AR and VR. Centres of excellence (CoEs) in this area will create thousands of job opportunities in the gaming and creative media spaces.

ML enables the machine to learn from the various tasks it performs to improve itself. AI ensures that the machine applies that learning algorithm best based on different situations. For example, a system could be trained using deep learning to recognise more complex scenarios or components. A camera's view of parts in an engine could suggest a repair to the technician, with instructions pinpointed on the image.

Also, an AR display that uses AI image recognition can streamline complex maintenance in aviation.

AI-based personalised shopping is another example that utilises VR devices.

AR and ML can be used together to solve business challenges and revolutionise everyday lives. AR harnesses the ability of ML to learn and remember imagery. Similarly, AR apps created for mobiles utilise AI and ML capabilities of smartphones.

Immersive head-mounted displays (HMDs) are primarily used to experience AR and VR. AR HMDs drive enterprise usage as hands-free tools for business process improvement and training.

Realty industry

Housing.com has launched a first-of-its-kind innovative VR-based solution called RealtyVision 2.0 for 3D presentation of real-estate projects. The portable product offers 3D walkthrough support, and can be controlled from both the buyer's as well as the seller's end. It reflects the consumer's real-time journey via touchscreen. This feature enables the seller to guide, explain and navigate during the digital tour through convenient remote/pointer-based interactions.

The state-of-the-art technological solution offering integrated 3D experience works without an Internet connection, easily facilitating customer visits. It allows the builder to sync all data through the cloud, and update fresh information and new projects as per need.

Ravi Bhushan, group CPTO, Housing.com, PropTiger.com and Makaan.com, says, "VR is moving from being a novelty to a necessity in the real-estate sector. Homebuyers, developers and real-estate agents are demanding a better and more efficient way of discovering and showcasing properties. We believe that this innovation will empower consumers to make smarter purchasing decisions more efficiently."

AR and VR for military training needs (Credit: www.hill.af.mil)

Manufacturing industry

3D AR and VR platforms have applications in multiple domains for developing real-time simulations that can create virtual environments. Such platforms are transforming the way OEMs execute designing, marketing, training, autonomous driving and human-machine interactions. These enable OEMs to innovate, deploy and test human-machine interfaces (HMIs) for the next generation of vehicles through a real-time 3D-simulated environment.

OEMs are witnessing the ability to conceptualise autonomy to advanced levels of safety and smartness without undergoing physical testing to keep their business costs down. Customers can also configure or experience in VR before physical cars are available in retail.

Security services

AR and VR can solve complex problems of the digitising world. With constant innovation and integration of AR/VR, Indian defence systems can enhance their strategy and use these technologies for such visual information as horizon line, aircraft speed and altitude. Anumukonda Ramesh, country head - India, Unity Technologies, says, "Defense and PSU sectors are one of the biggest users of simulation and visualisation for training, learning and product designing."

AR military systems can test any weaponry before use; for example, ARC4 connects to military satellites and drones as well as highlights dangers and threats in AR mode.

VR, AR and MR require new computing platform designs to replicate the